

13_January_2023 • Unit 2

1H + 2H source-order past paper

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

Contents

- 1H Q01 • Statistics Toolkit • 4 marks
- 1H Q03 • Bearings, Scale Drawing & Constructions • 2 marks
- 1H Q04 • Angles in Polygons & Parallel Lines • 3 marks
- 1H Q05 • Ratio Problem Solving • 3 marks
- 1H Q06 • Percentages • 7 marks
- 1H Q09 • Powers, Roots & Standard Form • 3 marks
- 1H Q10 • Solving Inequalities • 6 marks
- 1H Q11 • Graphing Inequalities • 3 marks
- 1H Q17 • Direct & Inverse Proportion • 3 marks
- 1H Q18 • Functions • 4 marks
- 1H Q20 • Differentiation • 5 marks
- 1H Q22 • Sequences • 4 marks
- 1H Q24 • Vectors • 5 marks
- 2H Q07 • Simultaneous Equations • 3 marks
- 2H Q08 • Compound Interest & Depreciation • 4 marks
- 2H Q12 • Cumulative Frequency Diagrams • 5 marks
- 2H Q14 • Circle Theorems • 5 marks
- 2H Q15 • Statistics Toolkit • 3 marks
- 2H Q17 • Algebraic Proof • 4 marks
- 2H Q19 • Area & Volume of Similar Shapes • 4 marks
- 2H Q20 • Simultaneous Equations • 5 marks
- 2H Q21 • Volume & Surface Area • 3 marks
- 2H Q22 • Solving Inequalities • 3 marks
- 2H Q24 • Graphs of Functions • 2 marks
- 2H Q25 • Functions • 4 marks

Answer ALL TWENTY FOUR questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1** 80 students entered a dancing competition.

The table gives information about the length of time, in minutes, for which each student spent dancing.

Time (m)	Frequency
$0 < m \leq 12$	11
$12 < m \leq 24$	25
$24 < m \leq 36$	23
$36 < m \leq 48$	15
$48 < m \leq 60$	6

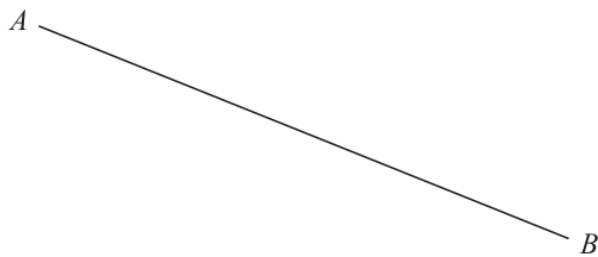
Work out an estimate for the mean length of time the students spent dancing.

..... minutes

(Total for Question 1 is 4 marks)

Q	Working	Answer	Mark	Notes
1	$6 \times 11 + 18 \times 25 + 30 \times 23 + 42 \times 15 + 54 \times 6$ $(= 2160)$ or $66 + 450 + 690 + 630 + 324 (= 2160)$ [lower bound products are: 0, 300, 552, 540, 288] [upper bound products are: 132, 600, 828, 720, 360]		4	M2 for at least 4 correct products added (need not be evaluated) or If not M2 then award: M1 for consistent use of value within interval (including end points) for at least 4 products which must be added or correct midpoints used for at least 4 products and not added
	"2160" $\div$ "80"			M1 dep on at least M1 Allow division by their Σf provided addition or total under column seen
	Correct answer scores full marks (unless from obvious incorrect working)	27		A1
				Total 4 marks

- 3 Use ruler and compasses only to construct the perpendicular bisector of line AB
You must show all your construction lines.



(Total for Question 3 is 2 marks)

3	Two pairs of intersecting arcs with equal radii centre A and B		2	M1 for arcs that intersect within or on the guidelines or correct perpendicular bisector without arcs.
	<i>Working required</i>	Bisector with construction arcs		A1 for a fully correct bisector with two intersecting arcs
				Total 2 marks

- 4 The diagram shows a pentagon.

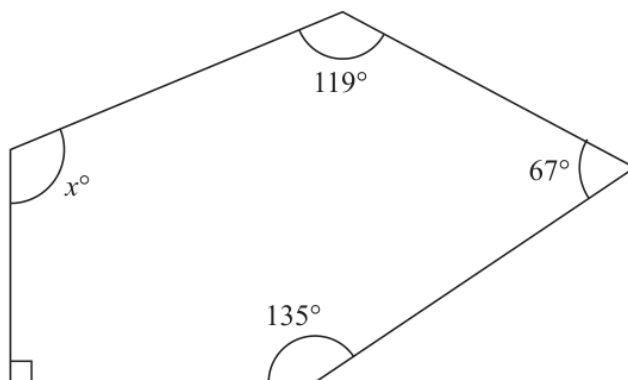


Diagram **NOT**
accurately drawn

Work out the value of x

$x =$

(Total for Question 4 is 3 marks)

4	$3 \times 180 (= 540)$ or $360 - [(180 - 90) + (180 - 135) + (180 - 67) + (180 - 119)] (= 51)$ or $360 - (90 + 45 + 113 + 61) (= 51)$		3	M1
	$90 + 135 + 67 + 119 + x = "540"$ oe $411 + x = "540"$ oe or $"540" - (90 + 135 + 67 + 119)$ or $3 \times 180 - (90 + 135 + 67 + 119)$ oe or $540 - 411$ or $180 - "51"$ oe			M1
	<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	129		A1
Total 3 marks				

- 5 In a box, there are only green sweets, orange sweets and yellow sweets.

There are 280 sweets in the box so that

the number of green sweets : the number of orange sweets = 2 : 3

and

the number of orange sweets : the number of yellow sweets = 1 : 5

Work out how many green sweets there are in the box.

(Total for Question 5 is 3 marks)

5	2 : 3 : 15 oe or 20 or (1 : 5) × 3 or (1 : 5 =) 3 : 15 or 2n : 3n : 15n e.g. 4 : 6 : 30 or G(reen) = 2, O(range) = 3, Y(ellow) = 15		3	M1
	$\frac{2}{20}$ ' 280 oe or 14 × 2 or $\frac{2}{2+3+15}$ ' 280 oe or $\frac{2n}{2n+3n+15n}$ ' 280 oe			M1
	Correct answer scores full marks (unless from obvious incorrect working)	28		A1 or 28 : 42 : 210 or 28 , 42 , 210 If not in this order must be labelled correctly
				Total 3 marks

6 Shane bought a car.

The amount Shane paid for the car was \$32 000

Theresa also bought a car.

To pay for this car, Theresa paid a deposit of \$18 000 together with 14 monthly payments of \$1160

Theresa paid more for her car than Shane paid for his car.

(a) Work out how much more Theresa paid as a percentage of the amount Shane paid.

..... %
(4)

Kylie bought a van.

After 1 year, the value of the van was \$39 865

During this year, the value of the van decreased by 15%

(b) Work out the value of the van when Kylie bought it.

\$.....
(3)

(Total for Question 6 is 7 marks)

6	(a)	18 000 + 14 × 1160 (= 34 240) oe or 18 000 + 16 240 (= 34 240)		4	M1
		"34 240" – 32 000 (= 2240) or $\frac{"34\ 240"}{32\ 000}$ (= 1.07)			M1
		$\frac{"2240"}{32\ 000}(\times 100)$ or $\frac{"34\ 240"}{32\ 000} \times 100 (= 107)$ or "1.07" – 1 (= 0.07)			M1
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	7		A1
	(b)	e.g. 1 – 0.15 (= 0.85) or 100(%) – 15(%) (= 85(%))		3	M1
		e.g. 39 865 ÷ 0.85 or 39 865 ÷ 85 × 100 oe			M1
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	46 900		A1
Total 7 marks					

- 9 (a) Write 6.25×10^{-4} as an ordinary number.

.....
(1)

- (b) Work out $(2.4 \times 10^{12}) \div (9.6 \times 10^4)$
Give your answer in standard form.

.....
(2)

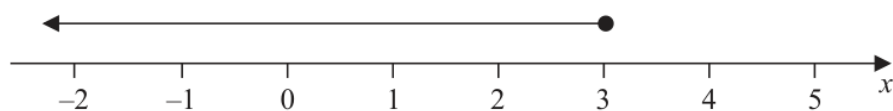
(Total for Question 9 is 3 marks)

9	(a)		0.000 625	1	B1
	(b)	25 000 000 oe e.g. 25×10^6 or 0.25×10^8 or 2.5×10^n $n \neq 7$		2	M1
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	2.5×10^7		A1
					Total 3 marks

10 (a) Factorise $y^2 - 2y - 48$

.....
(2)

(b) Write down the inequality shown on the number line



.....
(1)

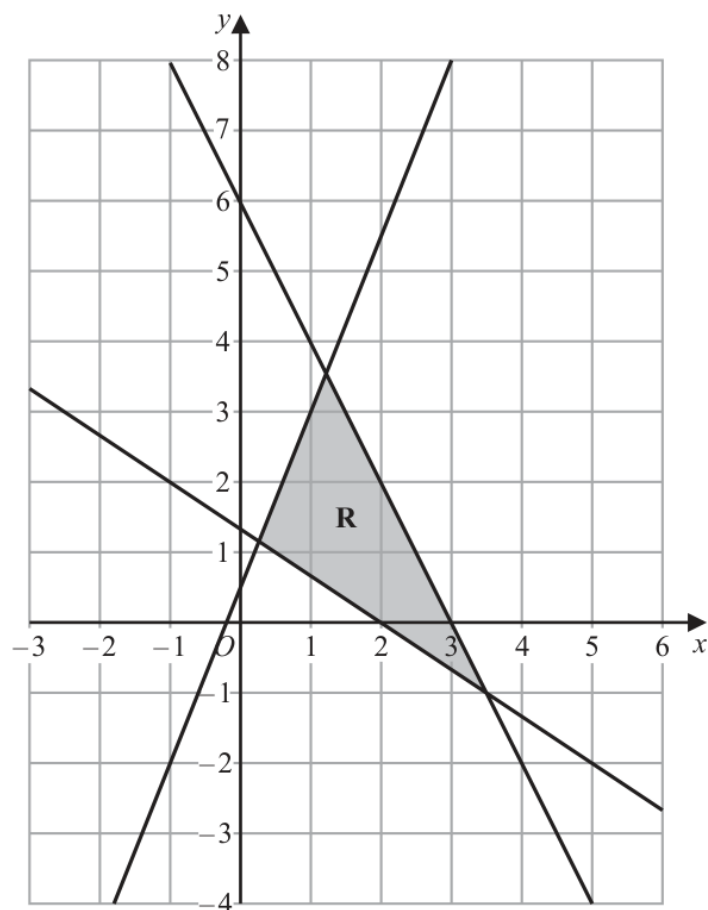
(c) Solve the inequality $7w + 6 > 12w + 14$

.....
(3)

(Total for Question 10 is 6 marks)

10	(a)	$(y \pm 6)(y \pm 8)$ or $y(y+6)-8(y+6)$ or $y(y-8)+6(y-8)$		2	M1 or for $(y \pm a)(y \pm b)$ where $ab = -48$ or $a + b = -2$
			$(y+6)(y-8)$		A1 oe Allow any letter for y
	(b)		$x \leq 3$	1	B1 allow $3 \geq x$ Allow any letter for x
	(c)	$6-14 > 12w-7w$ oe or $7w-12w > 14-6$ oe		3	M1 Condone = rather than > or any other sign for this mark.
		$-8 > 5w$ or $-5w > 8$ or $-w > \frac{8}{5}$ or $w > -\frac{8}{5}$ or $w = -\frac{8}{5}$ oe			M1 Condone = rather than > or any other sign for this mark.
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	$w < -\frac{8}{5}$		A1 oe accept $-\frac{8}{5} > w$ Must have correct sign on answer line dep on M1 (sight of correct answer in working space and just $(w =) -\frac{8}{5}$ oe on answer line gains M2 only)
					Total 6 marks

11



The region **R**, shown shaded in the diagram, is bounded by the straight lines with equations

$$2x + y = 6$$

$$2y = 5x + 1$$

$$3y + 2x = 4$$

Write down the three inequalities that define **R**

.....

.....

.....

(Total for Question 11 is 3 marks)

11		$2x + y \leq 6$ $2y \leq 5x + 1$ $3y + 2x \geq 4$	3	B3 oe for all three correct (B2 oe for any two correct) (B1 oe for any one correct) $2x + y \leq 6$ equivalent to $y \leq -2x + 6$ oe $2y \leq 5x + 1$ equivalent to $y \leq 2.5x + 0.5$ oe $3y + 2x \geq 4$ equivalent to $y \geq -\frac{2}{3}x + \frac{4}{3}$ oe Allow the following inequalities $2x + y < 6$ oe $2y < 5x + 1$ oe $3y + 2x > 4$ oe
				Total 3 marks

17 y is inversely proportional to $\sqrt{x}$

$y = c^4$ when $x = c^2$ where c is a positive constant.

Find a formula for y in terms of x and c
Give your answer in its simplest form.

(Total for Question 17 is 3 marks)

17	$y = \frac{k}{\sqrt{x}}$ or $ky = \frac{1}{\sqrt{x}}$ or $\sqrt{x} = \frac{k}{y}$ oe		3	M1 (NB. Not for $y = \frac{1}{\sqrt{x}}$) Constant of proportionality must be a symbol such as k (Allow c for k for this mark only)	M2 for $c^4 = \frac{k}{\sqrt{c^2}}$ oe
	$c^4 = \frac{k}{\sqrt{c^2}}$ oe or $k = c^4 \times \sqrt{c^2}$ oe			M1 for substitution of x and y into a correct formula	
	<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	$y = \frac{c^5}{\sqrt{x}}$		A1 oe e.g. $y = c^5 \times \frac{1}{\sqrt{x}}$ Award 3 marks if answer is $y = \frac{k}{\sqrt{x}}$ on the answer line and $k = c^5$ clearly given in the body of working of the script	
					Total 3 marks

18 The function f is such that $f(x) = \frac{k}{x}$ where $x \neq 0$ and k is an integer.

(a) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots \quad (1)$$

The function g is such that $g(x) = 2 - 3x^4$ where $x \neq 0$

The function h is such that $h(x) = \frac{3x}{2-x}$ where $x \neq 2$

(b) (i) Find $g(-2)$

$$\dots \quad (1)$$

(ii) Express the composite function hg in the form $hg(x) = \dots$
Give your answer in its simplest form.

$$hg(x) = \dots \quad (2)$$

(Total for Question 18 is 4 marks)

18	(a)		$\frac{k}{x}$	1	B1 allow kx^{-1}
	(b)(i)		-46	1	B1 cao
	(ii)	$\frac{3(2-3x^4)}{2-(2-3x^4)}$ or $\frac{6-9x^4}{2-2+3x^4}$ oe or $\frac{6-9x^4}{3x^4}$ oe		2	M1
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	$\frac{2-3x^4}{x^4}$		A1 allow $\frac{2}{x^4} - 3$ oe
					Total 4 marks

20 The radius of a right circular cylinder is x cm.

The height of the cylinder is $\left(\frac{800}{\pi x} - x\right)$ cm.

The volume of the cylinder is V cm³

Find the maximum value of V

Give your answer correct to the nearest whole number.



(Total for Question 20 is 5 marks)

20	$(V =) \pi x^2 \left(\frac{800}{\pi x} - x \right)$ or $800x - \pi x^3$ oe		5	M1 for volume of cylinder (in terms of one variable, e.g. x or r)
	$\left(\frac{dV}{dx} = \right) 800 - 3\pi x^2$			M1ft for differentiating an expression in one variable to find 800 or $\pm 3\pi x^2$ (must come from a cubic in the form $800x \pm \pi x^3$ or $800x \pm ax^3$ or $bx \pm \pi x^3$ where $a \neq 0$ and $b \neq 0$)
	" $800 - 3\pi x^2 = 0$ "			M1ft dep on previous M1 for equating their $\frac{dV}{dx}$ to zero (must be a quadratic in the form $800 \pm ax^2$ or $b \pm 3\pi x^2$ where $a \neq 0$ and $b \neq 0$)
	$(x =) \sqrt{\frac{800}{3\pi}} (= \sqrt{84.8(8263...)})$ or $\frac{\sqrt{9600\pi}}{6\pi}$ or $9.2(13177...)$			A1 for a correct value of x Allow use of quadratic formula
	<i>Award marks within the range from correct working</i>	4914		A1 accept 4910 – 4914
				Total 5 marks

- 22 The first term of an arithmetic series is $(2t + 1)$ where $t > 0$
The n th term of this arithmetic series is $(14t - 5)$

The common difference of the series is 3

The sum of the first n terms of the series can be written as $p(qt - 1)^r$ where p , q and r are integers.

Find the value of p , the value of q and the value of r
Show clear algebraic working.



$$p = \dots\dots\dots q = \dots\dots\dots r = \dots\dots\dots$$

(Total for Question 22 is 4 marks)

22	$2t + 1 + (n - 1)3 = 14t - 5$		4	M1 for the use of n th term $= a + (n - 1)d$ to find n
	$(n =) 4t - 1$ or $(n =) -1 + 4t$			A1
	$(S_n =) \frac{"4t-1"}{2} [2(2t+1) + ("4t-1"-1)3]$ or $(S_n =) \frac{"4t-1"}{2} [2t+1+14t-5]$ oe			M1 for the use of S_n formula (must be in terms of t) Allow their expression for n dep on M1
	Working required	$p = 2$ $q = 4$ $r = 2$		A1 dep on M2 allow $(S_n =) 2(4t-1)^2$ Values of p , q and r must come from correct working
				Total 4 marks

24 $OAED$ is a quadrilateral.

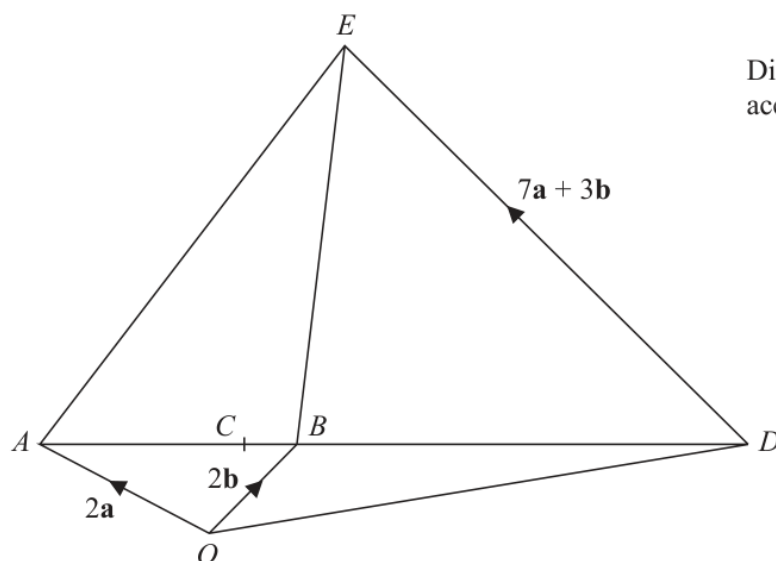


Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 2\mathbf{b} \quad \vec{DE} = 7\mathbf{a} + 3\mathbf{b}$$

$$AB:BD = 1:2$$

The point C on AB is such that OCE is a straight line.

Use a vector method to find the ratio of $OC:CE$



(Total for Question 24 is 5 marks)

24	e.g. $(\overrightarrow{AB}) = 2\mathbf{b} - 2\mathbf{a}$ oe or $(\overrightarrow{BA}) = 2\mathbf{a} - 2\mathbf{b}$ oe or $(\overrightarrow{BD}) = 2(2\mathbf{b} - 2\mathbf{a}) (= 4\mathbf{b} - 4\mathbf{a})$ oe or $(\overrightarrow{AD}) = 3(2\mathbf{b} - 2\mathbf{a}) (= 6\mathbf{b} - 6\mathbf{a})$ oe		5	M1
	e.g. $(\overrightarrow{OE}) = 2\mathbf{b} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= 3\mathbf{a} + 9\mathbf{b})$ oe or $(\overrightarrow{OC}) = 2\mathbf{a} + \lambda(2\mathbf{b} - 2\mathbf{a}) = (2 - 2\lambda)\mathbf{a} + 2\lambda\mathbf{b}$ oe or $2\mathbf{b} + \lambda(2\mathbf{a} - 2\mathbf{b})$ or $(\overrightarrow{CE}) = (2\mathbf{b} - 2\mathbf{a}) - \lambda(2\mathbf{b} - 2\mathbf{a}) + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} = (1 + 2\lambda)\mathbf{a} + (9 - 2\lambda)\mathbf{b}$			M2 for 2 correct paths seen M1 for 1 correct path seen Any correct path for OC passing through A or B involving a variable
	e.g. $\frac{2 - 2\lambda}{2\lambda} = \frac{1 + 2\lambda}{9 - 2\lambda}$ oe or $\frac{2 - 2\lambda}{2\lambda} = \frac{3}{9}$ oe or $\frac{(1 + 2\lambda)}{(9 - 2\lambda)} = \frac{1}{3}$ oe or $\lambda = \frac{3}{4}$ or $(2 - 2\lambda)\mathbf{a} + 2\lambda\mathbf{b} = \mu(3\mathbf{a} + 9\mathbf{b})$ or $\lambda = \frac{3}{4}$ or $\mu = \frac{1}{6}$ or $2\mathbf{b} + \lambda(2\mathbf{a} - 2\mathbf{b}) = \mu(3\mathbf{a} + 9\mathbf{b})$ or $\lambda = \frac{1}{4}$ or $\mu = \frac{5}{6}$			M1 for comparing coefficients of $\mathbf{a}$ and $\mathbf{b}$ for $(OC \text{ and } CE)$ or $(OC \text{ and } OE)$ or $(CE \text{ and } OE)$ OC is a multiple of OE Two different paths for OC
		1 : 5		A1 dep M2 oe e.g 2 : 10
	Working required			Total 5 marks

24 ALT	e.g. $(\overrightarrow{AB}) = 2\mathbf{b} - 2\mathbf{a}$ oe or $(\overrightarrow{BA}) = 2\mathbf{a} - 2\mathbf{b}$ oe or $(\overrightarrow{BD}) = 2(2\mathbf{b} - 2\mathbf{a}) (= 4\mathbf{b} - 4\mathbf{a})$ oe or $(\overrightarrow{AD}) = 3(2\mathbf{b} - 2\mathbf{a}) (= 6\mathbf{b} - 6\mathbf{a})$ oe		5	M1
	e.g. $(\overrightarrow{OE}) = 2\mathbf{b} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= 3\mathbf{a} + 9\mathbf{b})$ oe			M1
	e.g. $(\overrightarrow{AE}) = 2\mathbf{b} - 2\mathbf{a} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= \mathbf{a} + 9\mathbf{b})$ oe			M1
	$[\overrightarrow{AE} = \lambda \overrightarrow{AD} + \mu \overrightarrow{OE}]$ $\mathbf{a} + 9\mathbf{b} = \lambda(6\mathbf{b} - 6\mathbf{a}) + \mu(3\mathbf{a} + 9\mathbf{b})$ oe or $1 = -6\lambda + 3\mu$ oe and $9 = 6\lambda + 9\mu$ oe or or $\lambda = \frac{1}{4}$ or $\mu = \frac{5}{6}$			M1
		1 : 5		A1 dep on M2 oe e.g 2 : 10
	Working required			Total 5 marks

7 Solve the simultaneous equations

$$\begin{aligned}5x + 4y &= -2 \\ 2x - y &= 4.4\end{aligned}$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 7 is 3 marks)

Question	Working	Answer	Mark	Notes
7	eg $5x + 4y = -2$ $+ 8x - 4y = 17.6$ $(13x = 15.6)$ eg $[x = \frac{4.4 + y}{2}]$ oe $5\left(\frac{4.4 + y}{2}\right) + 4y = -2$ oe	eg $10x + 8y = -4$ $- 10x - 5y = 22$ $(13y = -26)$ eg $[y = 2x - 4.4]$ oe $5x + 4(2x - 4.4) = -2$ oe	3	M1 multiplication of one or both equation(s) with correct operation selected (allow one arithmetic error) (if + or - is not shown then assume it is the operation that at least 2 of the 3 terms have been calculated for) or correct rearrangement of one equation with substitution into second
	eg $5 \times "1.2" + 4y = -2$ or $2 \times "1.2" - y = 4.4$	eg $5x + 2 \times "-2" = 4.4$ or $2x - "-2" = 4.4$		M1 (dep on previous M1 but not on a correct first value) correct method to find second unknown – this could be a correct substitution into one of the equations given or calculated or starting again with the same style of working as for the first method mark
	Working required	$x = 1.2$ $y = -2$		A1 oe eg $x = \frac{6}{5}$ for both solutions dependent on first M1
Total 3 marks				

- 8 Matteo is going to invest 5000 Swiss francs for two years.

He can invest his money in Bank **G** or in Bank **H**.

Bank **G**
1.6% per year
compound interest

Bank **H**
2.9% interest added after
two years

The total amount of interest Matteo would receive at the end of two years from Bank **G** is more than the amount of interest Matteo would receive at the end of two years from Bank **H**.

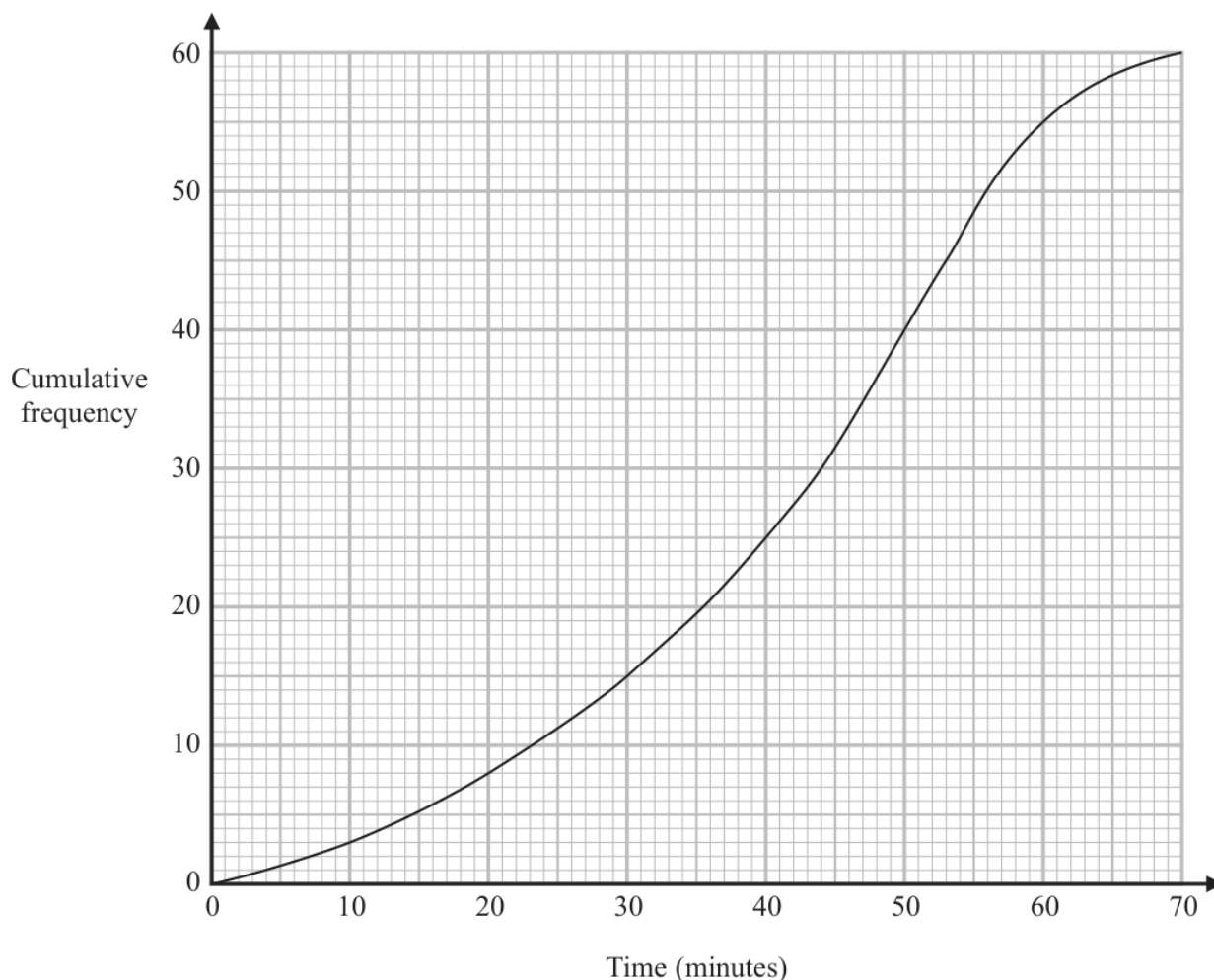
How much more?

..... Swiss francs

(Total for Question 8 is 4 marks)

Question	working	Answer	Mark	Notes
8	$\frac{2.9}{100} \times 5000 (= 145)$ oe or $1.029 \times 5000 (= 5145)$ oe or $1.029^2 \times 5000 (= 5294....)$ oe or $0.058 \times 5000 (= 290)$ oe or $1.058 \times 5000 (= 5290)$			M1 Bank H
	5000×0.016 oe (= 80) oe or 5000×1.016 oe (= 5080) oe or $5000 \times 0.032 (= 160)$ oe or $5000 \times 1.032 (= 5160)$ oe	M2 for 5000×1.016^2 (= 5161.28)	4	M1 Bank G
	$(80 + 5000) \times 0.016 (= 81.28)$ oe or $5080 \times 1.016 (= 5161.28)$ oe			M1 Bank G
	Correct answer scores full marks (unless from obvious incorrect working)	16.28		A1
Total 4 marks				

- 12 The cumulative frequency graph gives information about the time, in minutes, each of 60 people took to shop in a market.



- (a) Use the graph to find an estimate for the median time people took to shop in the market.

..... minutes
(1)

- (b) Use the graph to find an estimate for the number of people who took longer than 55 minutes to shop in the market.

.....
(2)



- (c) Use the graph to complete the frequency table to give information about the time, in minutes, each of the 60 people took to shop in the market.

Time taken to shop in the market (m minutes)	Frequency
$0 < m \leq 10$	3
$10 < m \leq 20$	5
$20 < m \leq 30$	
$30 < m \leq 40$	
$40 < m \leq 50$	
$50 < m \leq 60$	
$60 < m \leq 70$	

(2)

(Total for Question 12 is 5 marks)

Question	Working	Answer	Mark	Notes																
12 (a)		43.5 - 44.5	1	B1 ±0.5 small square																
(b)	eg reading of 48 - 49		2	M1 For correct method to start the question eg a vertical line from 55 up to the line and a horizontal line from the correct point on the curve or a mark on the curve at the correct point and a mark on the vertical axis at the correct point or a correct reading of 48 to 49																
	Correct answer scores full marks (unless from obvious incorrect working)	11 or 12		A1 Allow an answer of 11 or 12 (ie must be whole number)																
(c)	<table><tr><th>Time taken to shop in the market (m minutes)</th><th>Frequency</th></tr><tr><td>$0 < m \leq 10$</td><td>3</td></tr><tr><td>$10 < m \leq 20$</td><td>5</td></tr><tr><td>$20 < m \leq 30$</td><td>7</td></tr><tr><td>$30 < m \leq 40$</td><td>10</td></tr><tr><td>$40 < m \leq 50$</td><td>15</td></tr><tr><td>$50 < m \leq 60$</td><td>15</td></tr><tr><td>$60 < m \leq 70$</td><td>5</td></tr></table>	Time taken to shop in the market (m minutes)	Frequency	$0 < m \leq 10$	3	$10 < m \leq 20$	5	$20 < m \leq 30$	7	$30 < m \leq 40$	10	$40 < m \leq 50$	15	$50 < m \leq 60$	15	$60 < m \leq 70$	5		2	B2 All values correctly filled in (NB: first 2 are already completed) (B1 for 3 or 4 correct values from 7, 10, 15, 15, 5)
Time taken to shop in the market (m minutes)	Frequency																			
$0 < m \leq 10$	3																			
$10 < m \leq 20$	5																			
$20 < m \leq 30$	7																			
$30 < m \leq 40$	10																			
$40 < m \leq 50$	15																			
$50 < m \leq 60$	15																			
$60 < m \leq 70$	5																			
				Total 5 marks																

14 A, B, C and D are points on a circle, centre O

EBF is the tangent to the circle at B

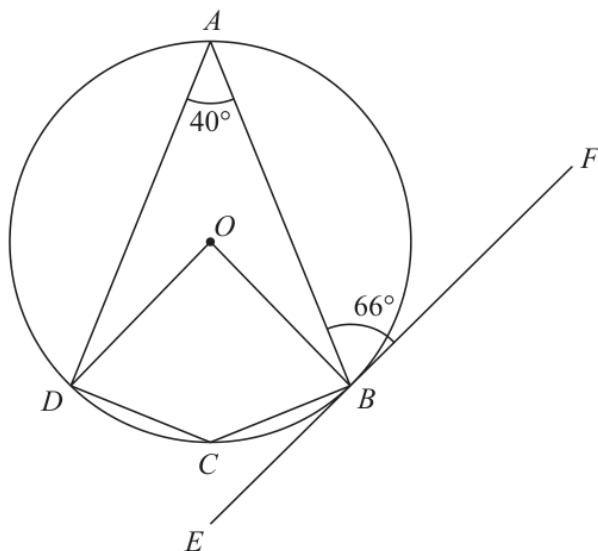


Diagram **NOT**
accurately drawn

(a) (i) Work out the size of angle DCB

o

(1)

(ii) Give a reason for your answer to (a)(i)

(1)

(b) Work out the size of angle ADO

o

(3)

(Total for Question 14 is 5 marks)

Question	Working	Answer	Mark	Notes
14 (a)(i)		140	1	B1
(a)(ii)		opposite angles of a cyclic quadrilateral (add to 180°) oe	1	B1 dep on B1 in (a)(i) or seeing 180 – 40 with no contradiction oe eg angle at centre is double (2 ×) angle at circumference oe AND angles around a point (or point 360)
(b)	$ADB = 66$ or $ABO = 90 - 66 (=24)$ or $BAO = 90 - 66 (=24)$ or $ODB = \frac{180-80}{2} (=50)$ or DOB reflex = 280		3	M1 Clearly labelled in working or shown on diagram
	For 2 of: $ADB = 66$ or $ABO = 90 - 66 (=24)$ or $BAO = 90 - 66 (=24)$ or $ODB = \frac{180-80}{2} (=50)$ DOB reflex = 280			M1 (award M2 for $360 - (280 + 40 + 24)$ oe
	Correct answer scores full marks (unless from obvious incorrect working)	16		A1
				Total 5 marks

- 15** Here is a list giving the numbers of runs scored last week by the eleven members of cricket team **A**.

2 3 4 6 21 26 27 32 34 61 72

The interquartile range of the numbers of runs scored last week by the eleven members of cricket team **B** was 42

Using a suitable calculation, write down one comparison between the numbers of runs scored by the members of cricket team **A** and the members of cricket team **B**.
Show your working clearly.

(Total for Question 15 is 3 marks)

Question	Working	Answer	Mark	Notes
15			3	M1 4 and 34 clearly indicated – either in list or in working (condone 26 also indicated in list)
				A1 For IQR for team A = $34 - 4 (= 30)$
		The IQR for Team B was higher than the IQR for Team A oe or Team B had an interquartile range of “12” more than team A or The runs scored were more spread out for Team B than for Team A oe or The runs for Team A were more consistent oe		B1ft Must fit dep on IQR stated for team A Either comparing the IQR correctly or for giving a comparison in context about spread as long as not contradicted by further statements as this would be choice NOT Team B scored more runs than team A The average score of B is higher than the average score of A The IQR of A was 30 while the IQR of B was 42 The range of B was more than the range of A
				Total 3 marks

- 17 Given that $8\sqrt{m} + \sqrt{49m} - \sqrt{9m} = k\sqrt{m}$
where k is an integer and m is a prime number,

(a) work out the value of k

$$k = \dots\dots\dots (1)$$

- (b) Show that $\frac{5 - \sqrt{18}}{1 - \sqrt{2}}$ can be written in the form $a + b\sqrt{2}$

where a and b are integers.

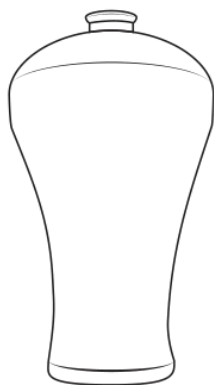
Show each stage of your working clearly.

(3)

(Total for Question 17 is 4 marks)

Question	WORKING	ANSWER	MARK	NOTES
17 (a)		12	1	B1
(b)	$\frac{5-\sqrt{18}}{1-\sqrt{2}} \times \frac{1+\sqrt{2}}{1+\sqrt{2}}$ or $\frac{5-\sqrt{18}}{1-\sqrt{2}} \times \frac{-1-\sqrt{2}}{-1-\sqrt{2}}$ oe		3	M1 Multiplying numerator and denominator by $1+\sqrt{2}$
	$\frac{5-\sqrt{36}+5\sqrt{2}-\sqrt{18}}{1+\sqrt{2}-\sqrt{2}-2}$ or $\frac{5-6-3\sqrt{2}+5\sqrt{2}}{-1}$ or $\frac{-5+6+3\sqrt{2}-5\sqrt{2}}{1}$ oe NB: allow $\sqrt{18}$ or $3\sqrt{2}$ $\sqrt{36}$ or 6 or $\sqrt{6}\sqrt{6}$			M1 Showing correct expansions (not necessarily as a fraction)
	working required	$1-2\sqrt{2}$		A1 dep on M2 (ie all stages of working must be shown convincingly) or for stating $a = 1$ and $b = -2$
Total 4 marks				

19 **A** and **B** are two similar vases.



A



B

Diagram **NOT**
accurately drawn

The vases are such that

$$\text{surface area of vase } \mathbf{B} = \frac{25}{64} \times \text{surface area of vase } \mathbf{A}$$

and that

$$\text{volume of vase } \mathbf{A} - \text{volume of vase } \mathbf{B} = 541.8 \text{ cm}^3$$

Calculate the volume of vase **B**

..... cm³

(Total for Question 19 is 4 marks)

Question	working	Answer	Mark	Notes
19	eg $\sqrt{\frac{25}{64}} \left(= \frac{5}{8} = 0.625 \right)$ or $\sqrt{\frac{64}{25}} \left(= \frac{8}{5} = 1.6 \right)$ or $\sqrt{25} : \sqrt{64} \text{ (5:8) or } \sqrt{64} : \sqrt{25} \text{ (8:5) or}$ $\frac{(\sqrt{25})^3}{(\sqrt{64})^3} = \left(\frac{125}{512} = 0.244140625 \right)$ oe or $\frac{512}{125} = 4.096$ $\frac{25^3}{64^3} = \frac{(\text{vol of B})^2}{(\text{vol of B} + 541.8)^2}$ or $\frac{25}{64} = \frac{(\text{vol of B})^{\frac{2}{3}}}{(\text{vol of B} + 541.8)^{\frac{2}{3}}}$ oe		4	M1 for a correct scale factor for length – may be given as a fraction or decimal or ratio or a correct scale factor for volume given as a fraction or decimal or ratio or a correct equation for the volume of vase B
	eg $B \left(\frac{512}{125} - 1 \right) = 541.8$ or $3.096B = 541.8$ oe or eg $A \left(1 - \frac{125}{512} \right) = 541.8$ or $\frac{387}{512} A = 541.8$ $(8^3)k - (5^3)k (= 387k) = 541.8$ or eg $(k =) \frac{541.8}{387} \left(= \frac{7}{5} \right)$			M1 For a correct equation for the volume of B or a correct equation for the volume of A
	eg (B) $541.8 \div \frac{387}{125}$ or $541.8 \div 3.096$ or eg $125 \times \frac{7}{5}$ or (A) $541.8 \div \frac{387}{512}$ (= 716.8) oe			M1 For a completely correct method to find the volume of vase B or vase A
	Correct answer scores full marks (unless from obvious incorrect working)	175		A1 cao
				Total 4 marks

20 Solve the simultaneous equations

$$\begin{aligned}y &= 7 - 2x \\ x^2 + y^2 &= 34\end{aligned}$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

20	$x^2 + (7 - 2x)^2 = 34$	$\left(\frac{7-y}{2}\right)^2 + y^2 = 34$	5	M1	substitution of linear equation into quadratic
	$5x^2 - 28x + 15 [= 0]$ oe	$5y^2 - 14y - 87 [= 0]$ oe		M1	dep on previous M1 for multiplying out and collecting terms, forming a three term quadratic in any form of $ax^2 + bx + c (= 0)$ where at least 2 coefficients (a or b or c) are correct and all are non-zero
	$(5x - 3)(x - 5) [= 0]$ or $\frac{-(-28) \pm \sqrt{(-28)^2 - 4 \times 5 \times 15}}{2 \times 5}$ or $5\left[\left(x - \frac{28}{10}\right)^2 - \frac{784}{100}\right] + 15 = 0$ oe or $x = 0.6$ and $x = 5$ (allow incorrect labels for x/y)	$(5y - 29)(y + 3) [= 0]$ or $\frac{-(-14) \pm \sqrt{(-14)^2 - 4 \times 5 \times (-87)}}{2 \times 5}$ or $5\left[\left(y - \frac{14}{10}\right)^2 - \frac{196}{100}\right] - 87 = 0$ oe or $y = 5.8$ and $y = -3$ (allow incorrect labels for x/y)		M1ft	dep on M1 for solving <i>their</i> 3 term quadratic equation using any correct method (if factorising, allow brackets which expanded give 2 out of 3 terms correct) (if using formula allow one sign error and some simplification – allow as far as $\frac{28 \pm \sqrt{784 - 300}}{10}$ or $\frac{14 \pm \sqrt{196 + 1740}}{10}$) (if completing the square allow as far as shown) or correct values for x or correct values for y dep on correct quadratic
	eg $y = 7 - 2 \times 5$ and $y = 7 - 2 \times 0.6$ (correct labels for x/y)	eg $5.8 = 7 - 2x$ and $-3 = 7 - 2x$ (correct labels for x/y)		M1ft	dep on previous M1 for substituting their 2 found values of x or y in a suitable equation or correct values for the other variable
	Working must be shown		$x = 0.6, y = 5.8$ $x = 5, y = -3$	A1	dep on M1 and the correct quadratic (allow coordinates) must be paired correctly

					Total 5 marks
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- 21** Given that the surface area of a sphere is $49\pi\text{cm}^2$
find the volume of the sphere.
Give your answer correct to the nearest integer.

..... cm^3

(Total for Question 21 is 3 marks)

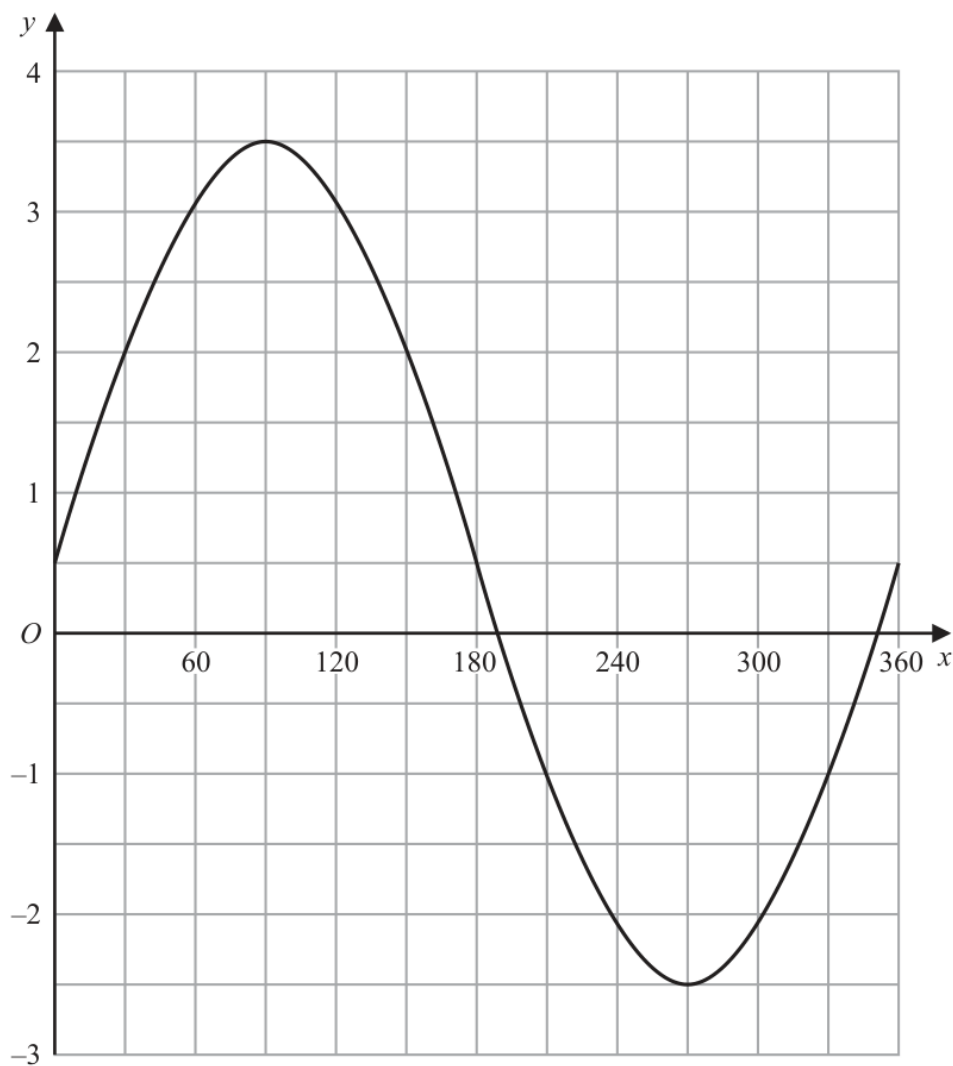
Question	Working	Answer	Mark	Notes
21	$r = \sqrt{\frac{49\pi}{4\pi}}$ oe (= 3.5)		3	M1
	[volume =] $\frac{4}{3} \times \pi \times 3.5^3$			M1
	<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	180		A1 awrt 180
				Total 3 marks

- 22** Solve the inequality $6x^2 + 37x \leq 35$
Show clear algebraic working.

(Total for Question 22 is 3 marks)

Question	Working	Answer	Mark	Notes
22	$(6x - 5)(x + 7)(=0) \text{ or }$ $\frac{-37 \pm \sqrt{37^2 - 4 \times 6 \times -35}}{2 \times 6}$ $6 \left[\left(x + \frac{37}{12} \right)^2 - \left(\frac{37}{12} \right)^2 \right] \dots \text{ oe}$		3	M1 A correct method to solve the quadratic equation $6x^2 + 37x - 35 (= 0)$ using any correct method (if factorising, allow brackets which expanded give 2 out of 3 terms correct) (if using formula allow one sign error in substitution and some simplification – allow as far as $\frac{-37 \pm \sqrt{1369 + 840}}{12}$) or completing the square as far as shown on left
	$\frac{5}{6}$ oe and -7			A1 dep on M1 correct critical values (allow 0.83...)
	<i>Working must be seen for both accuracy marks as asked for in question</i>	$-7 \leq x \leq \frac{5}{6}$		A1 dep on M1 oe eg $-7 \leq x \leq 0.83\dots$, $\left[-7, \frac{5}{6} \right]$ Accept $x \leq \frac{5}{6}, x \geq -7$
Total 3 marks				

24 The graph of $y = a \sin x^\circ + b$ is drawn on the grid.



Find the value of a and the value of b

$a =$

$b =$

(Total for Question 24 is 2 marks)

Q	working	Answer	mark	Notes
24		3	2	B1
		0.5		B1 oe
				Total 2 marks

25 The function f is such that $f(x) = 3x^2 - 12x + 7$ where $x \leq 2$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots\dots\dots$$

(Total for Question 25 is 4 marks)

Q	working	Answer	Mark	Notes
25	$y = 3(x^2 - 4x) + 7$ or $y = 3\left(x^2 - 4x + \frac{7}{3}\right)$ or $\frac{y-7}{3} = x^2 - 4x$ or $y = 3(x-2)^2 \dots$		4	M1 for a correct equation for a first step to complete the square
	eg $y = 3((x-2)^2 - 2^2) + 7$ or $y = 3((x-2)^2 - 2^2 + \frac{7}{3})$ or $y = 3(x-2)^2 - 5$ oe or			M1
	$(x-2)^2 = \frac{y+5}{3}$ oe eg $(x-2)^2 = \frac{y-7}{3} + 4$ or $x-2 = (\pm)\sqrt{\frac{y+5}{3}}$ oe			M1
	Correct answer scores full marks (unless from obvious incorrect working)	$2 - \sqrt{\frac{x+5}{3}}$		A1 oe NB: note only negative square root. Must be in terms of x any equivalent form
	Note: Allow candidates to swap x and y when finding the inverse			Total 4 marks
25 Alt	$3x^2 - 12x + (7-y) = 0$		4	M1 for a correct first step
	$(x =) \frac{12 \pm \sqrt{144 - 12(7-y)}}{6}$			M1
	$(x =) 2 \pm \sqrt{\frac{60+12y}{36}}$ oe			M1
	Correct answer scores full marks (unless from obvious incorrect working)	$2 - \sqrt{\frac{x+5}{3}}$		A1 oe NB: note only negative square root. Must be in terms of x any equivalent form
				Total 4 marks